

CHM111 Practice Exam on chapter(s): 9, 10

Name: _____

These questions assess your understanding of the material from Chapters 9, 10 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

1. Identify the main group to which X belongs, given the ionic compound $X_3(PO_4)_2$. Explain your answer.

ANSWER: _____

2. Draw the Lewis electron-dot symbol for an atom of P.

ANSWER: _____

3. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for carbon tetrafluoride.

ANSWER: _____

4. Which compound has a lower lattice energy, $BaCl_2$ or BaF_2 ? Explain your choice.

ANSWER: _____

5. Draw a valid Lewis structure for HCN.

ANSWER: _____

6. A particular boron atom has a total of two bonds to other atoms and has zero lone pairs of electrons around it. Calculate the formal charge on this boron atom.

ANSWER: _____

7. Rank the following elements from highest metallicty to lowest metallicty. Explain your ranking.

Cu, Ga, Au, Zn

ANSWER: _____

8. State the type of bonding (ionic, covalent, or metallic) that you would expect in H_2O . Explain your answer.

ANSWER: _____

9. Consider the following compounds: carbon disulfide, acetylene, carbon tetrachloride, and oxygen difluoride. Write the molecular formula for all of the listed compounds that have a dipole moment. If none have a dipole moment, write "None have a dipole moment". Explain your answer.

ANSWER: _____